

Portable sounds

- **What?** Walkman
- **Who?** Masaru Ibuka, Norio Ohga, Nobutoshi Kihara, Akio Monta, and Kozo Ohsonoe
- **Where and when?** Japan, 1979

In 1962, the Dutch electronics firm Philips introduced compact cassettes, which store music on magnetic tape (see p.205). The invention paved the way for the first truly portable music player, the Sony Walkman, in 1979. It was the brainchild of Sony co-founder Masaru Ibuka who wanted a way of listening to music on plane flights.



The device came with headphones for listening to music privately.

An early Sony Walkman, 1979

Released in 2001, Apple's iPod made the MP3 player globally popular



Users rotated a "click wheel" to browse the music stored on the device.

Compressed sounds

- **What?** MP3s
- **Who?** Karlheinz Brandenburg
- **Where and when?** Germany, 1989

Digital recordings take up a lot of computer memory. In 1989, a new technology called MP3 arrived. This significantly reduces an audio file's size by removing the parts people don't normally hear—a 40 MB song can be reduced to just 4 MB. The first MP3 players were released in 1999. Today, people can listen to their entire record collection on an MP3-playing app on their smartphone.

Wire-free sounds

- **What?** Bluetooth headphones
- **Who?** Several manufacturers
- **Where and when?** Various, 2002–2004

A problem with headphones is that they tether the user to the music source. People tried radio to overcome this, but it was not until the 2000s that Bluetooth, invented in the 1990s by the Swedish telecom giant

Ericsson, finally solved the problem. This radio-wave technology allows data to be transmitted over short distances, enabling people to enjoy wire-free music on their smartphones. The first wireless headphones were released in 2002.



Unlike vinyl records, CDs don't scratch easily.

Digital sounds

- **What?** Compact discs
- **Who?** Philips and Sony
- **Where and when?** Netherlands and Japan, 1982

In the 1980s and 1990s, compact discs (CDs) replaced vinyl records as the dominant format. Based on technology invented by the American James Russell in the 1960s, CDs store sound as digital information in pits molded on plastic disks, which can be read by a laser.

Headphones use radio waves to communicate with a smartphone.

